

CITS5508 Practice Exam — Chapter 8

Dimensionality Reduction

- This practice paper has **5 questions** for **60 marks**.
 - Allow approximately **two minutes per mark** ($\approx$ 2 hours total).
 - Only UWA-approved calculators permitted.
 - Answers are in the separate file *Practice Exam - Chapter 8 (Answer Key).md*.
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Question 1. (12 marks)

- (a) State **two** main motivations for reducing a dataset's dimensionality and **two** main drawbacks. (4 marks)
- (b) Explain the **curse of dimensionality**. Why does high-dimensional data tend to be **sparse**, and why does this increase the risk of **overfitting**? (4 marks)
- (c) Why is "just collect more data to fill the space" usually not a practical solution to the curse of dimensionality? (2 marks)
- (d) Once a dataset's dimensionality has been reduced (e.g. by PCA), can the operation be reversed exactly? Explain. (2 marks)
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Question 2. (12 marks)

- (a) Explain the difference between the two main approaches to dimensionality reduction: **projection** and **manifold learning**. (4 marks)
- (b) Briefly explain what a **manifold** and **manifold learning** mean, and state the **manifold assumption (hypothesis)**. (4 marks)
- (c) Using the Swiss roll as an example, explain why simple projection (e.g. dropping a dimension) can fail, and what manifold learning does instead. (4 marks)
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Question 3. (12 marks)

- (a) Explain how **PCA** chooses the axis (1st principal component) onto which it projects. Give the **two** equivalent justifications for this choice. (4 marks)
- (b) What is the **explained variance ratio**, and how would you use it to choose the number of dimensions to keep? (4 marks)
- (c) Write a scikit-learn snippet that applies PCA to keep **95%** of the variance, and a second line that prints how many components were actually retained. (4 marks)
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Question 4. (12 marks)

- (a) A 1,000-dimensional dataset is reduced with PCA, keeping an explained-variance ratio of 95%. How many dimensions will the resulting dataset have? Explain your reasoning. (3 marks)
- (b) Define **reconstruction error** in the context of PCA, and explain how it can be used for **anomaly detection**. (4 marks)
- (c) State when you would use each of: **regular PCA**, **incremental PCA**, **randomized PCA**, and **random projection**. (5 marks)
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Question 5. (12 marks)

(a) Explain how **random projection** reduces dimensionality, and why it can preserve distances reasonably well (refer to the Johnson-Lindenstrauss lemma). Does the required output dimensionality depend on the original number of features n ? *(5 marks)*

(b) **LLE** is a manifold-learning technique. Outline its two main steps at a high level. *(4 marks)*

(c) Name one dimensionality-reduction technique especially suited to **visualising clusters** in 2-D, and one **supervised** technique that projects onto the most class-discriminative axes. *(3 marks)*